Blockchain-based Proxy Re-Encryption Scheme for Data sharing in Internet of Vehicles

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***Abstract*—With the development of Internet of Vehicles(IoV) , the value of vehicle data has received more and more attention and it cannot be fully utilized without data circulation and sharing. However, due to problems such as data security, and low quality, the sharing of vehicular data grows slowly. This paper proposes a user-centric vehicular data sharing scheme based on consortium blockchain and proxy re-encryption to solve the problem of data silos among current vehicles, vehicular service providers, IoT devices, and other beneficiaries. Besides, we built a decentralized vehicular data market to help implement federated learning to achieve more accurate vehicle insurance pricing. During the training process, participants only exchanged intermediate calculation results without revealing other data to protect vehicular data privacy and improved the quality of smart vehicular services meantime. A large number of experiments and security analysis show that the scheme can achieve trusted identity authentication in data sharing of IoV, and effectively protect the user's data privacy, while at the same time reaching the actual business scenario-level time delay.**

***Index Terms*—blockchain, proxy re-encryption, data sharing, IoV, smart contract, federated learning**

I. Introduction

In the coming 5G era, the high-throughput and low-latency IoV will be marketed in large areas, and the rapid development of IoV represents the main trend of intelligent transportation[1]. IoV is not just connecting vehicles to vehicles, it is also connecting vehicles to pedestrians, vehicles to infrastructure (signal lights, etc.), vehicles to networks, vehicles to clouds. It plays an increasingly important role in improving the efficiency of transportation systems. At the same time, IoV will establish a new business model of vehicular service and improve the driving experience of vehicles, promote the development of autonomous driving technology, and provide vehicles owners with more intelligent and efficient driving services. A lot of vehicular data are saved by vehicle devices in the form of electronic records during the driving process. With the development of autonomous driving technology, big data, cloud computing, and other technologies, vehicular data has gradually become the focus of vehicles manufacturers and related companies in the automotive sector. For example, different users can collect driving data such as driving time, location, acceleration and deceleration, and turning through various onboard sensor devices and choose to share it to the related platforms for analysis and modeling, which is used to depict the user's driving risk level and serve as a reference for insurance pricing. Data sharing among vehicles or vehicles and other roles is conducive to improving the user's driving experience and the quality of intelligent transportation. Vehicular data has become an indispensable resource for the IoV application, and the sharing of vehicular data has become the consensus of the entire vehicular industry. However, due to the security and efficiency of data sharing, the sharing of vehicular data became extremely difficult at first. Many users can not share their data among different vehicles and data silos are formed. Moreover, the accumulation of vehicular data consumes a lot of resources and the value of vehicular data itself is not reflected.

Therefore, many vehicular data sharing solutions are trying to ensure the security of vehicular data while achieving high efficiency so that it can be shared among different vehicles to make full use of the value of vehicular data. In [2], the authors use the cloud-based data sharing solution to exchange vehicle information in IoV and introduce third-party auditors in the cloud to verify the integrity of shared data to improve the efficiency of audit tasks. A new mobile P2P architecture in the IoV is proposed in[3], which can make each vehicle node under the same driving test unit to share resources, reducing the bandwidth pressure on the server-side. These methods have a certain role in promoting the data sharing of the traditional IoV architecture, but the security issues in distributed vehicular network data sharing are still unresolved.

Blockchain is a decentralized distributed ledger technology that was applied in bitcoin [4]. In recent years, the blockchain has developed rapidly, and people have begun to apply it to the

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sharing of vehicular data [5]. Blockchain has non-tamperable and good cryptographic features to ensure that the data can be traced back. The smart contract of blockchain can be used as a pre-agreed program that can automatically execute once it is triggered[6]. Setting blockchain as the underlying infrastructure, each subject participating in vehicular data sharing acts as a blockchain node, and the data sharing operations between the nodes can be used as a record on the public ledger, which can be traced and reviewed and greatly facilitates accountability afterward. There have been many attempts[7]-[10] to share vehicular data sharing schemes based on blockchain, but many of them are limited to the conceptual level. Because of the use of blockchain technology in IoV, the performance of the solution must be considered first, such as latency and throughput, etc. Second, it depends on whether it helps improve the data security. Therefore, there are many issues to be considered in actual blockchain-based IoV data sharing applications.

In this paper, we propose a user-centric vehicular data privacy sharing scheme based on consortium blockchain [11]-[17] and proxy re-encryption [18], which is compatible with existing cloud-based data storage and combines the trusted decentralized identifiers mechanism (DID)[19] to solve data silos issues among current vehicles, vehicular service providers, IoT vehicular devices, etc. And we give the rights of controlling vehicular data sharing back to individual users. Besides, we also integrated federated learning[20] for joint modeling of private vehicular data. Our contributions can be generalized as follows:

- We propose a user-centric vehicular data privacy sharing scheme based on consortium blockchain, proxy re-encryption, and DID to solve the problem of trusted identity verification and data security in the process of vehicular data sharing, and also meet the performance requirements of actual application scenarios.
- We build a decentralized data AI market, providing federated learning suites for the participants of data sharing. It brings the AI model to the data of edge devices such as the On-Board Diagnostics(OBD)[21] and realizes federated machine learning under the protection of vehicular data privacy to improve the quality of Usage-Based Insurance(UBI)[22] service, which is greatly beneficial to the development of smart IoV.
- We use smart contracts to issue data sharing tasks in our scheme, which fully protects user data rights and interests. We also put forward discussions on data sharing development trends including laws and regulations, data rights, and data privacy computing technologies, which are also the key issues in the future IoV data sharing field.

The remainder of the paper is organized as follows: We systematically summarized the work related to vehicular data sharing in section II. In section III, we introduce some background concepts. In section IV, we present our overall architecture and the workflow for addressing the current challenges of vehicular data sharing will be introduced. Experimental details and results of our proposed scheme will be described in section V. A theoretical security and performance analysis is proposed in section VI. At the same time, we discussed the future development of data sharing in section VII. Finally, we conclude the paper in section VIII.

II. Related Work

The predecessors of data sharing in IoV have done a lot of work. Most of them mainly focus on two aspects, one is the performance of the data sharing system, and the other is the data security problem in the process of data sharing.

In [2], Runa used the cloud-based method to exchange vehicle information in IoV, vehicles can join different interest groups to form a social network where they can facilitate the modification or sharing of data in the cloud. Kong Q proposed the improved Parlier cryptosystem to achieve sensory data aggregation and preserve location privacy, thereby enabling data sharing in IoV[23]. Garg Sahil proposed "Sec-IoV", a multi-stage model for anomaly detection, to ensure vehicle-to-vehicle (V2V) communication in IoV settings, to explore solutions to address the security risks of highly dynamic deformation and distributed characteristics of the vehicle network[24]. Fan proposed a two-way authentication protocol to protect the driver's privacy, using transparent computing in the protocol and client hardware to improve the quality of the user experience[25].

As blockchain technology became popular, people began to combine blockchain technology in IoV to ensure the security of data sharing. In the current research of vehicular data sharing based on blockchain, the common idea is to design a data sharing architecture purely based on the blockchain. Another approach is to integrate some cryptographic methods based on the blockchain, such as building a trusted execution environment to ensure the security of the data sharing environment.

In [26], the authors proposed a quality-driven data sharing mechanism based on the consortium blockchain, which ensures trust in on-chain and off-chain data and expressed the data sharing issue as a reverse auction. By using an auction model driven by data quality, the data quality assessed by expectation maximization can be used to ensure trust in off-chain data. [27] proposed a credibility framework called Vcash for identifying denial of traffic services to solve the problem of trust in the Internet application level. Each vehicle directly communicates with RSU for traffic incident propagation and verification, and shares verified traffic data in the Vcash system.In [28],the authors developed a lightweight vehicle-mounted blockchain security (LVBS) data sharing framework in drone-assisted IoV, and developed a credit-based consensus algorithm in the blockchain to track the behavior of the nodes and record transactions data to improve the efficiency and security of reaching consensus. They also used reinforcement learning algorithms to optimize the pricing and quality of data sharing strategies for data contributors and data consumers. In [29], the authors proposed a novel blockchain-based architecture named BlockAPP for vehicle authentication and privacy protection and seamless access control to IoV. The proposed architecture supports optional traceability features. The authors proposed an

IoV data sharing scheme that rewards a certain number of cryptocurrencies for vehicles that contribute to blocking generation to improve the quality of the data and the degree of activity of the system in [30]. This is mainly to promote IoV data sharing from economic means.

Based on the blockchain, some studies have also integrated other privacy protection technologies to ensure the security of data sharing. In [31],Nick proposes a decentralized proprietary data market named Sterling which builds a reliable execution environment for the use of data for the training of machine learning models. However, this method requires the use of hardware, which has a relatively high-cost requirement and has proved that there is a side-channel attack against TEE [32].

In [33], a homomorphic encryption-based data sharing framework based on chaotic mapping is proposed. Under this framework, a new type of reversible data hiding (RDH) based on lesion extraction to protect privacy and improve data quality, but homomorphic encryption implements fewer operations and is difficult to support for some complex calculations and cost of implementation is very high. [34] builds a secure data market named BlockMarkchain based on the blockchain, each data seller can exchange authenticated data with the buyer in a secure environment, but there may be malicious behavior and lack of effective verification. In [35], Hao Guo proposes a hybrid architecture of blockchain, and edge nodes based on an attribute-based multi-signature scheme to promote data management without revealing sensitive information. The method with attribute-based multi-signature increases system overhead, which is not very suitable for the data sharing scenarios that require high-time performance.

In recent years, some new blockchain-based privacy computing technologies such as zero-knowledge proof[36], and multi-party secure computing[37] have also emerged. The integration of AI methods into IoV data sharing is also a very common method. In [38], the authors proposed a security solution using reinforcement learning (RL) technology to prevent jamming attacks in Mobile edge caching (MEC) systems, using lightweight authentication and a secure collaborative caching scheme to protect Data privacy. By integrating AI and blockchain into the wireless network to achieve flexible and secure resource sharing, a safe and intelligent architecture for the next generation wireless network is proposed in [39]. As is known to all, practical applications must consider technical feasibility, efficiency, cost, and other factors. To choose a relatively high-performance solution, we are inspired by NuCypher's proxy re-encryption solution[40] and try to use encrypted access control performed through a decentralized network to move beyond the limitation of using a consensus network to safely store and process private encrypted data. The solution supports and provides a secure infrastructure for decentralized applications to share sensitive data varying from vehicular data sharing to identity management. At the same time, we considered integrating federated learning in vehicular data sharing to solve security problems in vehicular data sharing, collecting vehicles, and driving behavior data through IoV devices, conducting multi-party joint modeling through machine learning models. The output parameters of the model are normalized and finally converted into user driving behavior scores to help improve UBI and other automotive-related services.

III. Preliminaries

In this section, some preliminaries used in our blockchain-based data sharing scheme are introduced.

A. *Consortium Blockchain*

Consortium blockchain refers to a blockchain with several organizations in management and it only allows the institutions within the system to read, write data, and send transactions. Due to the limited nodes, the consortium blockchain has the advantages of faster transaction speed, stronger access control, lower transaction cost, etc. It is easier to seek common ground while reserving differences in terms of interests. [41] proposed a consortium blockchain framework to solve the problem of detecting malicious code in malware and extracting corresponding evidence in mobile devices. Compared with the previously released algorithm, the new method can achieve faster detection speed. Therefore, the consortium blockchain is more in line with enterprise-level requirements in IoV and the application is more targeted and efficient. At the same time, the consortium blockchain can also maximize the data management characteristics of the blockchain technology in the enterprise application market, reducing the trust cost of transactions in different units' non-trust environments in data sharing scenarios.

B. *Smart Contract*

The smart contract concept was first proposed in the early 1990s by computer scientist, lawyer, and cryptographer Nick Szabo [42]. A smart contract is a computer protocol designed to propagate, verify, or execute contracts in an informative manner. It allows trusted transactions without third parties, which are traceable and irreversible. It is a pioneering practice to write smart contracts in the digital form in the blockchain. The characteristics of the blockchain technology ensure that the entire process of smart contract storage, reading, and execution is transparent and traceable, and cannot be tampered with. Therefore, the smart contract on the blockchain can be automatically executed once triggered, and no one can tamper with it. At the same time, a set of state machine systems are built by the consensus algorithm built in the blockchain, so that the smart contract can run efficiently, not only can support programmable contracts and can be decentralized, the process is transparent and traceable. In the IoV data sharing scenario, users only need to issue smart contracts to initiate data sharing transactions, reducing the cost of user communication, and at the same time building a trust foundation among data sharing participants.

C. *Proxy Re-encryption*

Proxy re-encryption is to entrust a trusted third party or a semi-honest proxy to convert the ciphertext encrypted by the public key of one party into a ciphertext that can be unlocked by the another party's private key to implementing key sharing[43].

In the classic proxy re-encryption scenario, there are several roles. Alice(sk_A, pk_A) is the data owner and she wants to transfer data M to Bob(sk_B, pk_B) with semi-honest proxy (such as the company providing cloud services) between them. The specific communication process is:

Alice: Encrypt the plaintext M with her public key. The M is the content that A wants to send to B.

$$C_{pkA}=encrypt(pk_A,M) \tag{1}$$

Alice: Send the ciphertext C_{pkA} generated by equation (1) to the semi-honest Proxy and generate the conversion key $rk_{A\text{->}B}$ for it. This key is generated by A for the Proxy.

$$rk_{A\text{->}B}=rekey(sk_A,pk_B) \tag{2}$$

Proxy: The re-encryption key generated by A ($rk_{A\text{->}B}$) in equation (2) converts the ciphertext C_{pkA} into the ciphertext that can be decrypted by Bob's private key. The Proxy only provides the calculation conversion service and it can't obtain the plaintext and sends the generated ciphertext C_B to Bob.

$$C_B = reencrypt(rk_{A\text{->}B}, C_{pkA}) \tag{3}$$

Bob: Decrypted C_B generated by equation (3) with sk_B to get the plaintext M that A wants to secretly share.

D. Distributed Identity(DID)

In the real world, there are a variety of data used to describe the identity of entities and relationships between entities, such as ID cards, driving licenses, etc. Distributed identity(DID) is a distributed identity protocol [44], which can be implemented based on the existing blockchain. The real entity (person or object) can obtain a unique identity on the blockchain through DID. Through DID, a credible electronic credential can be generated for users, which can be verified by the institution after authorization by users, thereby simplifying business processes and reducing the risk of private data leakage. DID can establish a trusted identity and it is widely used in vehicular data sharing affairs.

E. OBD and UBI

On-Board Diagnostics(OBD) is an extended detection system for automobile fault diagnosis. OBD conducts real-time monitoring of vehicle and its components such as engines, catalytic converters, particulate traps, oxygen sensors, emission control systems, fuel systems, etc. to access or deal with different fault information.

UBI model vehicle insurance (Usage-Based Insurance) is a kind of quantifiable insurance based on the combination of the driving behavior of the owner and the use of vehicle-related data. Real-time monitoring of vehicle data such as mileage and fuel consumption through OBD vehicle-mounted intelligent terminals, combined with driving behavior data such as the number of "three urgency" and violations of owners, through the processing of big data technology to assess the risk level of the driving behavior of the owner, through the risk level index for each The owner provides a customized policy, and the premium depends on the comprehensive consideration of the owner's actual mileage, driving time, driving location, specific driving behavior, and other indicators. In fact, at the end of the 20th century, there was a UBI product based on driving behavior, which collected driving data such as driving time, location, acceleration and deceleration, and turning through various onboard sensing devices, and passed it to the central platform for analysis and modeling, which was used to depict the driving risk of users as a reference for insurance pricing, but its data security issues are highly questioned.

F. Federated Learning

Federated learning does not use the data required for the aggregation model training to perform centralized calculations, but rather distributes the computation of the machine learning to the distributed computing of the participating parties' databases. To coordinate the nodes of each distributed computing, the gradient-related data needs to be transmitted, which requires privacy protection of the gradient-related data to prevent the original data from being inversely derived. The entire model training process does not transmit any raw data, thus protecting the privacy of the data owners [45]. The self-owned data of each participant in the federated learning process exists locally and does not need to be taken out. The method can establish a virtual common model through the parameters exchange method under the encryption mechanism. When building a virtual model, the data itself does not move, nor does it reveal privacy and affect data compliance. In this way, the built models serve only local targets in their respective regions. Each participant has the same identity and status under such a federated mechanism. Federated learning enables two or more parties to use data in cooperation, it can be used to solve the problem of data silos.

IV. PROPOSED SCHEME

A. Scheme Layers

When designing a blockchain system, we usually must consider several levels of factors. When we abstract the contents of the specific protocols and operations involved in the scheme, it is conducive to understanding the workflow of the real vehicular data sharing system. The whole scheme is mainly divided into several layers as shown in Fig. 1, and the functions of each layer are introduced as follows:

a) Access layer: Devices in the vehicular data access layer include mobile devices, wearable devices, etc. Vehicular data can be transmitted to vehicular nodes through these devices, and the data owners can trace the usage of the data.

b) Data layer: The data layer mainly involves DID authorization and encryption algorithms such as proxy re-encryption in the data sharing process.

c) Storage layer: The scheme we proposed is compatible with existing technical solutions. Vehicular data storage can use existing cloud service plans or be stored on the blockchain. The use of data requires the authorization of the users to be cross-vehicles sharing.

d) Network layer: The network layer mainly involves the P2P network, the mechanism of data transmission, and verification to ensure the safety of the data transmission process.

e) Reward layer: It mainly includes an economic incentive issuance system and distribution system. Its function is to provide certain incentives, encourage nodes in IoV to participate in the security verification work in the blockchain, and incorporate economic factors into the blockchain technology system to encourage compliance with the rules. IoV nodes involved in accounting will be rewarded, and those who do not follow the rules will be punished. In actual application scenarios, the transportation department can join the major vehicular manufacturers to establish a unified standard token mechanism, which is conducive to promoting data sharing.

f) Sharing layer: The sharing layer of the scheme mainly involves the participants of vehicular data sharing and specific applications such as the IoV service providers, vehicles owners, and relevant beneficiaries. Data sharing requires the participants to actively participate and coordinate, clarify the interests of all parties, and finally reach a consensus. At the same time, vehicles data sharing also requires the supervision, management, and guidance of relevant departments such as the government's big data management department in the transportation system. In the next section, we will describe the workflow of the proposed scheme in detail.

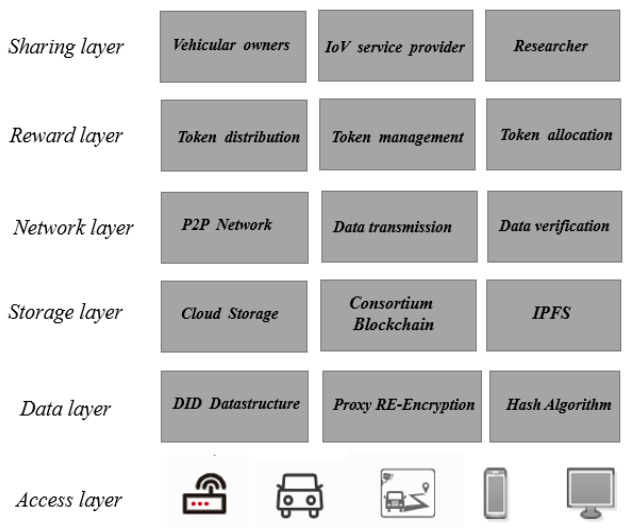

Fig. 1. The layers design of our proposed scheme

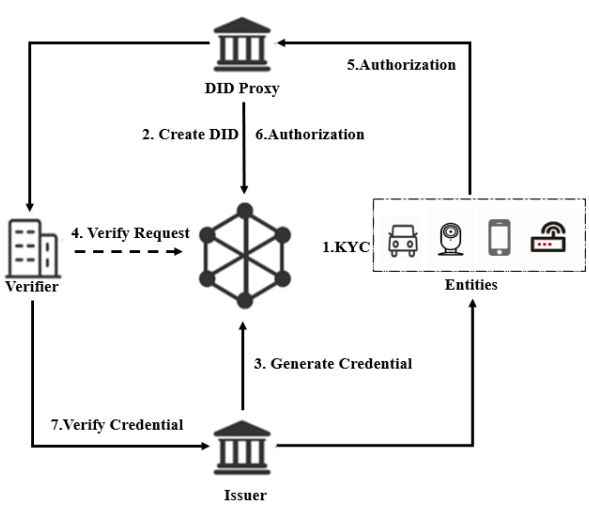

Fig. 2. Trusted identity mechanism:DID

B. *Trusted identity authentication*

We use the DID mechanism to ensure the trustworthiness of the vehicular data sharing participants including vehicles, IoT equipment, vehicular service providers, mobile devices, etc. It can minimize some malicious attacks such as Sybil attack[46] and Eclipse attack[47]. Its workflow is mainly as shown in Fig. 2.

- *Step 1*. The entities participating in vehicular data sharing need to register the related information (KYC) and authorize the DID proxy to process the DID related service. The KYC information includes the name, role type, and related Identity information to support unique index in the proposed system.
- *Step 2*. DID Proxy creates a unique DID on the blockchain for the entity and deposits the authorization of entities. Here we choose the DID generation method by multiple hashes and then use base58 to unify the length of the DID sequence. This generation method is similar to the generation of bitcoin addresses [4] to ensure the uniqueness of the DID.
- *Step 3*. The issuer uses the DID to generate the credential and uploads the hash of certificate credential to the blockchain. Issuer generates the signature with an asymmetric encryption algorithm.
- *Step 4*. When an entity needs to use a certain IoV service, the service provider verifies entity's identity, and the service provider acts as a DID verifier. The verifier needs to request access to data on the blockchain and get authorization and then verifies the DID credential and the certificate content through the hash of blockchain and electronic signature. The verifier can verify the identity by verifying whether the hash digest of the data applicant's DID is consistent with the digest in the obtained digital credential.
- *Step 5*. The entity will authorize the service provider for verification if it needs to use the IoV service.
- *Step 6*. DID Proxy authorizes the verification.
- *Step 7*. Verifier verifies the credential and confirms entities' identification information.

The detailed algorithm description is shown in Algorithm 1:

Algorithm 1 Implementation of DID authentication

Input: DID_i, DID's credential, E_i's authorization
Output: The Effectiveness of DID_i
1. Entity(E_i):Select $name_i$, $role_i$,generate (Sk_i,Pk_i)
2. Proxy: Generates a random sequence S,E_i's KYC (K),a fix number N
   i. H_0 = Hash(S+K)
   ii. for i =1;i<= N do
      H_i = Hash(H_0)
   iii. DID_i = base58(H_i)
3. Issuer: generate $credential_i$ of DID_i
   i. Calculte Hash(DID_i) and $Signature_i$(Hash(DID_i), Sk_i)
   ii. Create $digest_i$ in blockchain;
4. Verifier: Given the announced DID_i , E_i's authorization, Issuer's Pk
   i. Decryption $credential_i$ and get $Signature_i$,E_i's Pk_i
   ii. Decryption $Signature_i$ and get the digest H_i
   iii. Calculte Hash(DID_i)
   iv. Verifying H_i == Hash(DID_i)
return True

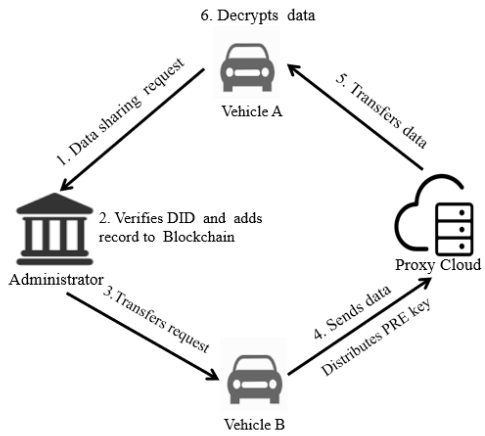

Fig. 3. Access vehicle data across vehicles

C. Data sharing across vehicles

After the process of DID setup in section A, a vehicle owner can apply for vehicular data sharing in the proposed scheme when he or she wants to access another vehicular data across vehicles. The workflow is mainly as shown in Fig. 3, in the following process, I regard the human behavior and the behavior of the vehicle as a whole because the data sharing between the vehicles still depends on the people to operate:

- *Step 1*: If a vehicle has a key pair (sk_{VA}, pk_{VA}) wants to view another vehicle(sk_{VB}, pk_{VB}) 's data across vehicles, it needs to send a data sharing request to the system administrator. The method of vehicle 's request is mainly through the initiation of a smart contract for data sharing. The contract specifies the data information, the data type and the subject, and the corresponding benefit division. The execution of the contract requires the consent of the subject to which the data belongs.
- *Step 2*: After verifying the DID of the vehicle that requests data, the system administrator adds the data sharing record to blockchain for easy auditing and traceability. Only the hash value of the data sharing transaction is stored on the blockchain and no specific data is stored.
- *Step 3*: The system administrator forwards the data sharing request to the data owner node. The nodes in the same channel can communicate with each other. When the administrator forwards the request to the vehicle, the administrator will first lookup the previous routing table. If there is no information about the target node, the request will be forwarded to the data owner through the P2P broadcast. We set a response time in our scheme, if this threshold is exceeded then this request will be abandoned. Through this method, the system overhead is reduced to some extent.
- *Step 4:*The data owner will encrypt the requested data with the private key sk_{VB}, send the ciphertext to his/her data storage location such as the public cloud platform if he/she agrees to the request and distribute the re-encrypted key($rk_{VB\text{->}C}$) to the proxy re-encryption cloud. Regardless of whether the data sharing is authorized by the vehicle user, the administrator similarly adds the record of the user's processing request to the blockchain. Then the task of data sharing will be carried out according to the provisions of the smart contract. The smart contract is executed on the blockchain, and the data applicant cannot access the data outside the contract. When the data is obtained, the contract will be automatically destroyed, leaving the corresponding transaction record for auditing.
- *Step 5*:After obtaining the data owner's authorization to share specified data and the re-encrypted key, the cloud proxy will query the data if the data owner has uploaded data to the cloud before and re-encrypt the data encrypted by data owner with re-encrypted key($rk_{VB\text{->}C}$). Then the proxy will transmit the data to the vehicle applying for data sharing.
- Step 6:Finally, the data usage vehicle decrypts the data owner's data by using his/her private key sk_{VA}. The detailed description of implementation is shown in Algorithm 2:

Algorithm 2 Implementation of Data sharing across vehicles

Input: $Vehicle_A$'s DID_A, (Sk_{VA}, Pk_{VA}),request of data,$Vehicle_B$'s DID_B
Output: The requested data,M
1. Implementation of Algorithm 1 with DID_A, DID_B
2. Administrator: verifies the request, send it to data owner vehicle B
3. Vehicle B:generates reencryption key
   i. KeyGen():a $\in Z_q$, (Sk_{VB},Pk_{VB})=(a,g^a)
   ii. ReKeyGen(Sk_{VB},Pk_{VA},N,t): Sk_{VB},Pk_{VA},a number of fragments N,a threshold t
   Computes N fragments of the reencryption key,each of them named kFrag
   iii. Send $rk_{VB->C}$ to cloud
4. Cloud: Given $rk_{VB->C}$, C_{VB}=Encryption(Pk_{VB},M)
   i. C_{VA} = ReEncryption($rk_{VB->C}$, C_{VB})
5. Vehicle A: M = Decryption(C_{VA}, Sk_{VA})
return M

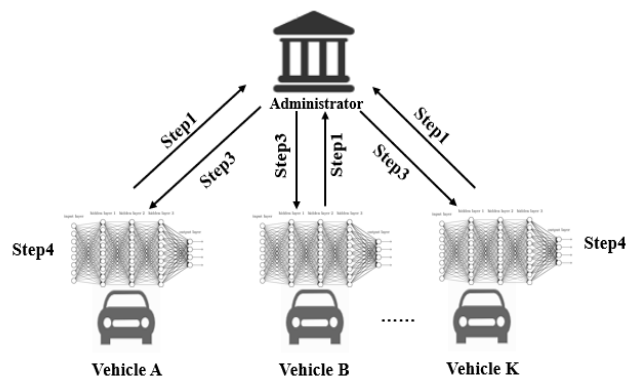

Fig. 4. Blockchain-based federated learning architecture. Any node in the consortium blockchain system that calls for federated learning needs can initiate data sharing transactions by issuing smart contracts to improve the quality of AI IoV services.

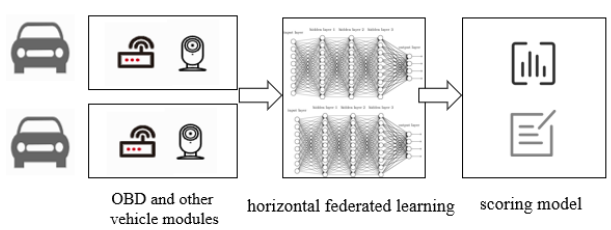

Fig. 5. A federated learning module based on OBD data. After the training results are normalized, a score will be obtained for UBI pricing reference

D. Vehicular data AI market

The design of the vehicular data market is as follows. The participants that want to obtain data for AI machine training also need to obtain data owners' authorization to share data as in Section B. The two parties initiate a smart contract for vehicular data joint modeling training, which stipulates the participants and benefit distribution plans for machine training according to market pricing rules and need to apply for a reliable third party such as a technical supporter designated by the regulatory department as the role of data aggregation. In our scheme, blockchain is used for data sharing proof storage and traceability, and related training is performed off-chain to improve the efficiency and accuracy of AI training. Before machine learning, the data is pre-processed by default. Here we only explain the process of AI joint training as shown in Fig. 4:

- *Step 1*: The administrator verifies the participants' DIDs and conducts federated learning with the users' consent on the blockchain. In the IoV scenario, vehicles owners can obtain data describing the users' driving behavior through vehicular sensor devices such as OBD for federated learning. The type of federated learning is determined according to the needs of the participants and the data. We mainly conduct horizontal federated learning on the data between multiple in-vehicle devices. Participants calculate the training gradient locally, use asymmetric encryption techniques to disguise the selected gradient, and send the masked result to the server. In the proposed scheme, the administrator acts as the role of the server. The administrator also needs to upload the hash digest of the training process to the blockchain for easy forensics.
- *Step 2*: The administrator performs a safe aggregation without knowing any participant's information. The effectiveness of aggregated data depends on operations similar to homomorphic encryption.
- *Step 3*: The administrator sends the aggregated results to the participants.
- *Step 4*: Participants update their respective models with the decrypted gradient. By iterating the above steps until the loss function converges, thus completing the entire training process, all participants will share the final model parameters. The detailed description of algorithm implementation is shown in Algorithm 3(Here take two participants as an example):

Algorithm 3 Implementation of vehicular data AI market

Input: A's DID_A, B's DID_B, model parameter D_A,D_B, (Sk_A,Pk_A), (Sk_B,Pk_B)
Output: The update of D_A, D_B
Step1. A: C_A = Encryption(Pk_A,D_A)
$result_A$ = mask(C_A)
B: C_B = Encryption(Pk_B,D_B)
$result_B$ = mask(C_B)
Administrator: verifies DID_A,DID_B, record in blockchain, receives $result_A$,$result_B$
Step2. SETUP: intermediate results $result_A$,$result_B$
Administrator: $result_0$ = aggregation($result_A$,$result_B$)
Step3. Administrator: send $result_0$ to A,B
Step4. A:unmask, decryption,updates D_A
B: unmask,decryption,updates D_B
return new $D_A^{'}$,new $D_B^{'}$

On this basis, Fig. 5 describes the federated learning mechanism based on OBD data proposed in the proposed scheme, vehicles owners can collect vehicle and driver behavior data in OBD and other vehicular devices to conduct the horizontal federated learning as the Figure 4 and the logistic regression (Logistic Regression) model is often used. The output parameters of the model will be normalized by Softmax function and finally converted into user driving behavior scores. This score is used as a reference for UBI pricing along with traditional pricing methods. This method greatly protects the user's data privacy, and at the same time can optimize the insurance pricing method. It can also implement the recommended report on car maintenance and improve the users' driving quality.

TABLE I
DID DATA STRUCTURES SAMPLE

FIELD	Description
Name	Vehicle AAA
DID	123456789abcdefghi#keys-1
Hash	86de8973acfe17d2f835956cef92624
Type	Vehicle/Data consumer
Credential serial	73ce2b24c5b0c60d3fb-01
Issue Date	Year/Month/Day
expire Date	Year/Month/Day
Proxy	XXX Proxy
Issuer	XXX administration Department

TABLE II
DATA SHARING SMART CONTRACT SAMPLE

FIELD	Description
struct datainfo	
description	The task description
datainfo	The data information
format	The format of data
totalSize	The size of data sharing
struct beneficiaries	
data_user	The user of data
data_owne	The owner of data sharing
storage	The storage of data
auditor	The auditor of data sharing
function data_request	The function describes the operation of the data request
function data_verify	The function describes the data sharing participants authentication such as calling DID verification interface,etc
function data_ sharing	The function describes the operation function of data sharing such as the method of data sharing and the length of data usage time,etc

V. EXPERIMENTAL DESIGN

A. Experimental setup

Firstly, we present the experimental analysis for our proof of scheme implementation using Fisco Bcos [48] consortium Blockchain. We deployed some nodes in a group on a Linux

machine(RAM>256G, Storage capacity>30T) to simulate the roles of the proposed system. This consortium blockchain provides a console for developers, which can directly deploy smart contracts in nodes. We deployed 8 nodes in two groups. One group contains four nodes for cross- vehicles data access, and the other group contains four nodes for initiating AI joint modeling training tasks. The blockchain nodes we use can join or exit at any time, but all nodes need to comply with the corresponding registration mechanism. Nodes on blockchain can initiate data sharing transactions through smart contracts. The transactions reach consensus through the PBFT consensus mechanism of the consortium blockchain and eventually are recorded on the blockchain for index.

Given the above-built blockchain, we set up the DID as described in Table 1 and tested the process of DID application. Table 1 shows the data structure of DID in the vehicular data sharing system. It includes the name of the data sharing participant, DID sequence, DID hash, user's type, credential serial number, issuance, and expiration date, and DID proxy, credential issuance agency. Through the DID mechanism, the identity of the entity is uniquely identified, which is easy to use for elated verification.

As for proxy re-encryption, we used pyUmbral[49],the reference implementation of Umbral, which is a proxy re-encryption network to empower privacy in decentralized systems. To generate an asymmetric key pair, we set SECP256K1[50]as a default curve to perform any ECC operations. When a node wants to grant another node access to open his or her encrypted messages, he or she creates re-encryption key fragments, or "kfrags", which are next sent to N proxies. In this experiment, we set different fragments and observe the performance of proxy re-encryption by controlling the size of the file.

The smart contract is written in solidity language[51], we deployed on the test environment in the consortium blockchain used, and the bottom layer uses the EVM module to support the execution and invocation of the smart contract. We deployed two contracts in the experiment. One contract is a vehicle's data sharing request contract. The contract specifies the relevant information of the data sharing task, the beneficiaries of data sharing, the scope of the data, and the corresponding operations. The specific field descriptions are shown in table 2. The other one is a smart contract for AI federated machine learning, which is roughly similar to the first smart contract.

In federated machine learning among nodes in the scheme, we used PySyft[52] and Pytorch[53] to perform the federated learning training CNN model experiment. PySyft is a framework that supports safe, private calculations in deep learning models. To explore the possibility of AI joint modeling in the blockchain system more objectively and succinctly, the dataset we used this time is the OBD-II dataset in Kaggle[54]. At the same time, we expanded some data with common driving behavior. Our final data set includes 16 characteristics such as the number of braking times per unit time, the number of sharp turns per unit time, the number of sharp accelerations per unit time, the fuel consumption per hour, the speed per hour, etc. The dataset contains 50000 examples for training and 10,000 examples for testing. We divide the data set into two users of different vehicles for federated learning, and we use the Logistic regression model to train the data set. We measured the three indicators of average loss, training time and accuracy rate to compare the results of federated learning and non- federated learning in our proposed system, so as to judge the merits of aligning samples between vehicles for federated learning.

B. Experimental result

The file storage of this DID is only a few hundred bytes, and the time consumption for creating and indexing is in the level of milliseconds. The Fig. 6 shows the time required for DID creation, credential issuance, verification request, user authorization, and verification. Overall, this process meets the speed requirements of the actual vehicular data sharing business.

In Fig. 7, we measured the time required to deploy a data sharing task smart contract. We used the kit provided by BCOS to deploy the contract on the consortium blockchain, which supports direct compilation. In the case of stable network communication, the time to deploy a contract is related to the file size of the contract, which is still insignificant relative to the overhead of other systems. Any authenticated user who wants to access data only needs to issue a smart contract, which greatly simplifies the process of data sharing.

In Fig. 8, we measured the time consumption of data transfer in four nodes' data sharing system using proxy re-encryption for different file sizes after the user authorization. At the same time, we also measured the average time consumption of file transfer in the case of different fragments. When the number of slices is fixed, with the gradual increase of shared data files, the time consumed also gradually increases, but on the whole, users still accept this delay time scale, it does not affect the availability of the system. In actual data sharing, if a large file is divided into multiple data pieces for transmission, its rate will gradually increase, and our experiments have proved this. The speed of encryption operations on the blockchain is also a hot topic that the industry has been discussing. As a result, the encryption algorithm selected is related to the performance of the blockchain. Here we have not dealt with too many challenges and only verified the feasibility of our proposed system.

Besides, in Fig. 9 we also present the analysis of the federated learning experiment on the proposed system. In this process, we achieved the safe aggregation of data. Compared with non-federated learning training, after the same first few epochs instead of the final ones, we found that the accuracy rate of federated learning is 2% to 3% higher than non-federated learning on the test set. But at the same time, the time consumption of federated learning is more than twice the non-federated learning's, and in some specific scenarios, it is worth the time to achieve this effect. More options for joint machine training implemented in the actual blockchain are training under the blockchain and the data sharing records during the training will be stored on the blockchain for traceability and forensics. The blockchain more is to implement a trust mechanism, through which distributed machine learning can be taken into application in more scenes.

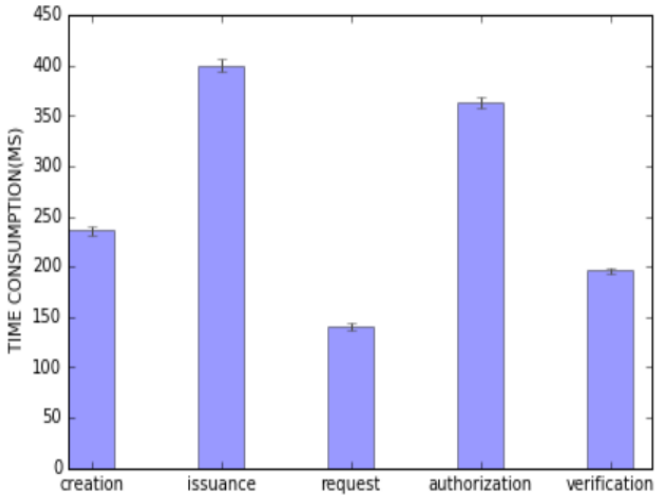

Fig. 6. Time consumption of DID setup in the proposed system based on the consortium blockchain. Normally, the KYC and DID setup is completed before the data sharing, which does not affect the data sharing process

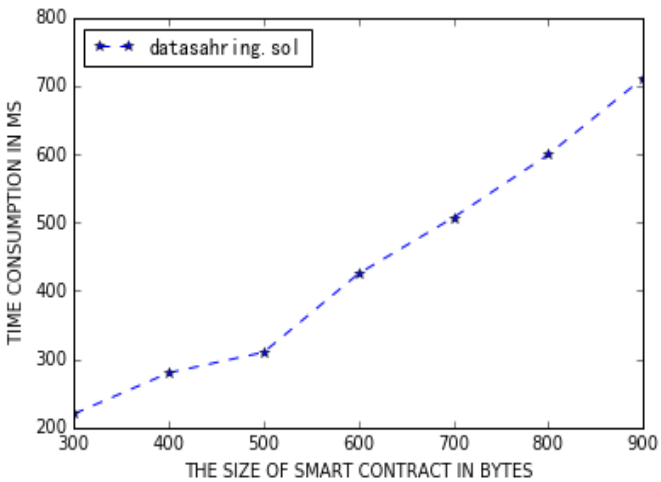

Fig. 7. Time consumption of DID setup on the consortium blockchain, it can be seen from above that the time consumption of the system to share data by issuing smart contracts conforms to the requirements of the IoV scenario

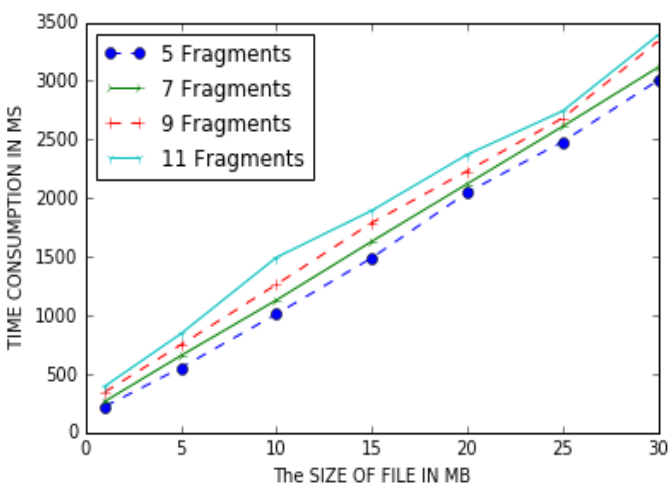

Fig. 8. Time consumption of sharing files based on proxy re-encrypting in the proposed system based on the consortium blockchain

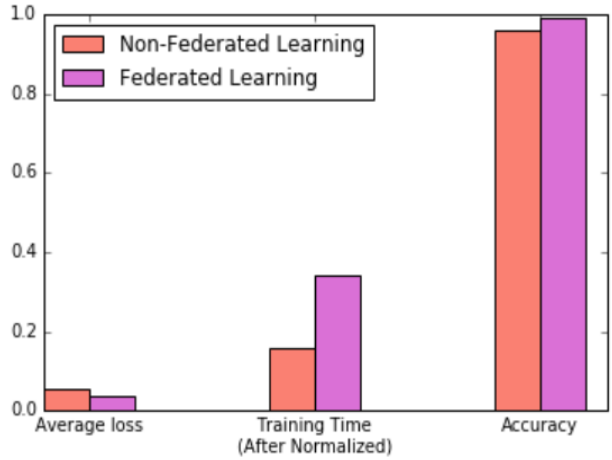

Fig. 9. Comparison of Federated Learning in and Non-Federated Learning In the proposed system

VI. Security Analysis

In this part, we analyze the performance of the proposed scheme in terms of the basic performance of blockchain, privacy protection, and efficiency.

A. *Tamper-proof and non-repudiation*

The scheme is based on the blockchain technology, so all information in it is non-tamperable because all nodes of the system jointly maintain the same public ledger. The blockchain stores the overall hash value of vehicles' data, vehicles' data authorization credentials, protocols for AI training usage data, etc., which can be traced and reviewed to facilitate vehicles owners' rights protection and return the rights of controlling data to vehicles owners.

As a trusted identity mechanism, DID is designed in the proposed scheme to achieve access to trusted entities. Each entity participating in data sharing has a unique identifier on the blockchain to avoid existing security attacks to a certain extent.

B. *Privacy protection*

- Anonymity: As with most blockchain systems, the proposed scheme will have an anonymous address for each vehicle, which is associated with the vehicle's key. All data sharing transactions in the scheme are anonymous and can greatly protect the privacy of vehicles owners.
- Privacy protection of sharing data: When accessing vehicle data across vehicles, it is a must to request vehicles' authorization. Proxy re-encryption is used to ensure data transmission security, which is compatible with existing IoV solutions using cloud service storage.
- Privacy protection for data federated learning: Federated learning is used in the proposed scheme to enable vehicular data to power smart IoV while ensuring users' data privacy. Compared to current solutions on the market, the privacy protection schemes we propose have a lot of advantages.

C. Efficiency

The scheme is based on the consortium blockchain. The transaction speed is much faster than the traditional public blockchain. On the Internet of Vehicles scenario, nodes can be dynamically joined or verified, and the size of a certain area is not particularly large. Secondly, data sharing is dynamic or time-sensitive, which is different from the traditional network architecture and causes less network overhead. And redundancy is not particularly large, the system administrator can choose to eliminate some malicious nodes or nodes with invalid identity, which is conducive to the construction of the entire data sharing ecology.

Secondly, we decouple the use and storage of data and put the process of federated learning and data sharing off the blockchain. There are only authentication and records on the blockchain, which has greatly improved the efficiency and speed of vehicular data sharing. Besides, the time of smart contract deployment and the overhead of proxy re-encryption meet the business needs of the actual data sharing scenes.

VII. Discussion

The rise of the data economy has made data increasingly important as an element of wealth distribution. But frustratingly, due to various reasons, a large amount of users' private data is sold every day in anonymous networks such as darknet. In the process of data sharing, we are worried about the value of the data itself, and the privacy and security of the data. The evaluation of data value needs to be determined according to specific application scenarios. With the emergence of new technologies such as blockchain, many new explorations in data sharing combined with new trust mechanisms have gradually taken place. The blockchain can make the asset attributes of the information itself become a digital asset, and this equity carrier can carry out value transfer on the chain, and the equity can include ownership, dividend rights, and so on. There are two main application scenarios for future data sharing affairs: one is pure information certification and traceability. The second is the process of further commercialization of information to commercialize the rights and transactions it carries. In the future, with the development of 5G technology, more and more edge computing devices and collaborative networks [55] composed of many IoT devices will provide a large amount of data, and data sharing between them will become a very common demand. At the right time, the blockchain provides a computing path where data can be shared but not disclosed, including through multi-party secure computing, federated learning, etc., which makes data exchange has new feasibility. Therefore, we think that there are several perspectives worth paying attention to in the future:

- Legislation and regulations on data security: the role of participating in data sharing needs to access data under the corresponding regulations, and at the same time we need to assign corresponding responsibilities according to the roles of data participants, so as to facilitate the responsibility clarification and trace. Regarding the scope of private data, most of the first impressions are related to individual users, such as identity, whereabouts, purchase behavior, etc., but the scope of private data is not limited to this, any data that does not want to be publicly disclosed is essential, such as internal business The data, corresponding to the owner of the data ownership, is correspondingly the data owner and should be protected.
- Data ownership rights and benefits distribution: Data has multiple dimensions of technical characteristics, but the economic characteristics of data are more complex. Data can generate value, so it has asset attributes. The data has the characteristics of both goods and services. Many data are public goods and can be freely used, modified, and shared by anyone for any purpose. Because most of the data is non-competitive, fewer data pertains to private products and public resources. The rise of data ownership protection awareness, coupled with the improvement of privacy protection legislation and the evolution of privacy data control from concentration to differentiation, for the existing business model with data as the main driving force, what will await a series of major challenges. The key to solving the challenge is not the zero-sum game between the data owner and the data user, but how to align the interest demands between the two to jointly achieve the optimal benefit.
- Related technologies for data privacy computing: cryptographic technologies such as verifiable computing, trusted computing, homomorphic encryption, and secure multi-party computing support data validation, making it possible to trade data usage rights without affecting data ownership, thereby constructing the property rights foundation of data transactions. Other privacy protection methods, such as access control technology in IoT devices[56] and supporting users to revoke or delete their data[57],etc. will gradually become the demands of data owners in the future. Because the subject that determines the usage of specific data is no more suitable than the owner of the data, which is also one of the characteristics of users' data sovereignty awareness. Blockchain technology is used as an infrastructure for the establishment and transmission of trust and plays an important role in data sharing and collaboration. Consequently, blockchain-based privacy computing technology and the security technology of the blockchain itself]will be a hot topic in the future and deserves our continuous attention and research.

VIII. Conclusion

In this paper, we propose a user-centric vehicular data sharing scheme based on consortium blockchain and proxy re-encryption. This scheme can realize vehicular data sharing across vehicles, and a decentralized vehicular data AI market is designed in the scheme. It can help IoV service providers, vehicles, and other related organizations to participate in federated machine learning, which is conducive to the development of IoV technology. We verified the proposed scheme through experiments and made relevant experimental analysis. In the end, we discuss the key issues and future trends of data sharing to bring some inspiration for future research. The most important thing is that the implementation of the scheme does great help to the existing vehicular data sharing problem solving and it can protect the privacy of users' data, which is also the original intention of scheme design.

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